OR laws

A	B	$Y = A+B$
0	0	$Y = 0+0 = 0$
0	1	$Y = 0+1 = 1$
1	0	$Y = 1+0 = 1$
1	1	$Y = 1+1 = 1$

The OR laws can be realised as

1st law	$A + 0 = A$
2nd law	$A + 1 = 1$
3rd law	$A + A = A$
4th law	$A + \overline{A} = 1$

AND laws

A	B	$Y = A.B$
0	0	$Y = 0.0 = 0$
0	1	$Y = 0.1 = 0$
1	0	$Y = 1.0 = 0$
1	1	$Y = 1.1 = 1$

The AND laws can be realised as

1st law	$A \,.\, 0 = 0$
2nd law	$A \,.\, 1 = A$
3rd law	$A \,.\, A = A$
4th law	$A \,.\, \overline{A} = 0$

The Boolean operations obey the following laws.

Commutative laws

$$A + B = B + A$$
$$A \,.\, B = B \,.\, A$$

Associative laws

$$A + (B + C) = (A + B) + C$$
$$A \,.\, (B \,.\, C) = (A \,.B) \,.\, C$$

Distributive laws

$$A(B + C) = AB + AC$$
$$A + BC = (A + B)\,(A + C)$$

The above laws are used to simplify complicated expressions and to simplify the logic circuitry.

10.7 DE MORGAN'S THEOREM

10.7.1 De Morgan's First Theorem

Statement

The first theorem states that the complement of the sum of two logical inputs is equal to the product of its complements.

Proof

The Boolean equation for NOR gate is $Y = \overline{A + B}$.

The Boolean equation for a bubbled AND gate is $Y = \overline{A} \,.\, \overline{B}$.

Both cases generate same outputs for same inputs. It can be verified using the following truth table.

A	B	$A+B$	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \,.\, \overline{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

From the above truth table, we can conclude $\overline{A+B} = \overline{A} \cdot \overline{B}$.

Thus De Morgan's first theorem is proved. Hence, a NOR gate is equal to a bubbled AND gate.

The corresponding logic circuit diagram is shown in Figure 10.47.

XII_Physics_Vol-2_UNIT-10_Part 1_EM.indd 231 19-12-2022 16:39:18